

Assignment 4 - Computational neuroscience

Group: Iva Papukchieva, Yan Radivilov, Rubén Rejón Lozano, Ole Rotteveel

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1 Introduction

1.1 Background

In the field of computational neuroscience, neural networks are considered one of the most compatible approaches for brain modelling. This is largely motivated by connectionism, where networks of interconnected artificial units are used to model aspects of neural computation. Each artificial neuron receives information from other units, combines these inputs, and produces an activation that is passed through the network.

This assignment explores three different neural network architectures and investigates how well their internal representations align with real brain data from the superior temporal gyrus (STG), a region of the auditory cortex [1] associated with auditory processing. The models introduce different architectural choices and levels of complexity, allowing us to investigate whether training, architecture, and model depth influence their alignment with brain representations. As a comparison metric we perform Representational Similarity Analysis (RSA) [2] between the models and the human STG response.

1.2 Data and Models

For this study we used the Santoro auditory dataset. This dataset comprises 288 auditory stimuli of one second in 6 distinct natural sound categories with their corresponding STG fMRI responses from 5 healthy human subjects. These responses are represented by a vector of fMRI response values across the STG voxels, where each element reflects the measured response of one voxel to that sound. Each voxel reflects the mass activity of many neurons. We compared three custom neural network models, namely a Waveform model, an Uninspired model and an Inspired model. We also analyzed YAMNet, which is a pretrained neural network for audio-event classification

The Waveform model operates directly on the waveform and uses Multilayer Perceptrons (MLP) transformations, the Uninspired model is a spectrogram based convolution model without trying to mirror biological auditory representation. The Inspired model adds recurrence to integrate information over time, an architecture inspired by auditory regions.

For each of our custom models we both analyzed them pretraining with randomly assigned weights and post-training with optimized weights and to examine the effect of architecture design.

1.3 Purpose and Research Questions

In this study we want to answer the following research questions:

1. Does training the custom models cause the models to align closer to the STG data?
2. Does our custom model architecture make a difference in how fast and if the custom models align to the real STG data?
3. Do the custom models become more aligned with the STG data with increased model depth?
4. Does the general model YAMNet outperform our custom trained models?
5. Does UMAP visualization help in determining the differences in the underlying topological learned structure between the models and the STG structure?

1.4 Study Type

As illustrated in Figure 1, the study follows a , by comparing different neural network models on the same brain-aligned auditory processing task.

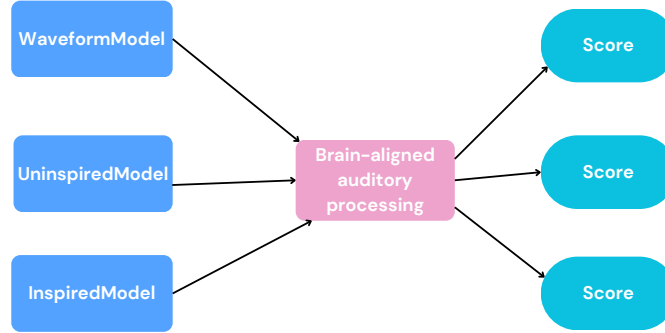


Figure 1: Contrastive study design. The three neural network models are compared on the same task of brain-aligned auditory processing, with their representational alignment to STG evaluated using RSA.

2 Methods

2.1 Dataset

The Santoro dataset comprises 288 natural auditory stimuli that are all 1 second long. The corresponding brain data consists of fMRI responses in the STG. The original dataset shows the responses to each sound using fMRI beta responses, which show the measure of the response associated with each stimulus at each voxel, where a voxel represents the mass activity of many neurons.

For the analysis the responses to the stimuli were represented by a vector containing the fMRI responses across the STG voxels included in the dataset. These responses were used as the representative responses to compare to the model representations.

We ensured that brain responses and model responses to auditory stimuli was consistent by keeping the order of stimuli the same between the dataset and the stimuli presented to the models. This ensures that the response and activation corresponding to the same stimuli is in the same position in their respective matrices.

The Waveform model processed the auditory waveforms directly, while for the Uninspired and Inspired models the stimuli were transformed to waveforms.

Model activations were extracted from multiple layers of each model. Each layer produced a representation of 288 auditory stimuli that could be compared with the corresponding STG brain representations. For every model the activations were extracted. The following representation were used to calculate the representational dissimilarity matrix between the models for the RSA.

2.2 Models

The way the audio stimuli were presented to the models differed based on their architecture. The Waveform model takes raw waveforms as input. It then flattens it and processes it through five fully connected layers with tanh activations, which is then followed by a linear classifier. The Uninspired model takes spectrograms as inputs and consists of three convolutional layers with

tanh activations, followed by two fully connected layers and then a linear classifier. The Inspired model also takes spectrograms as inputs. It has ReLu6 activations in its three convolutional layers to extract local time-frequency features. It then has two stacked recurrent (GRU) layers that integrate information over time followed by a linear classifier. The pretrained YAMNet model was trained on the Audioset-Youtube Corpus and predicts 521 different audio event classes. The model uses MobileNetv1 architecture based on depthwise-separable convolutional layers.

2.3 Representational Similarity Analysis

RSA Analysis: Step-by-Step

Step 1: Extract model activations

For every sound, extract its activation pattern at each layer of the model. Each sound is therefore represented by a vector of activations.

Step 2: Calculate pairwise distances

Within each layer, compute the Euclidean distance between the activation patterns of every pair of sounds:

$$d_{ij}^{(l)} = \left\| \mathbf{a}_i^{(l)} - \mathbf{a}_j^{(l)} \right\|_2$$

A small distance means that the model represents the two sounds similarly, whereas a large distance means that their representations are more different.

Step 3: Construct a model RDM

Arrange all pairwise distances into a Representational Dissimilarity Matrix (RDM):

$$\mathbf{D}^{(l)} = \begin{bmatrix} 0 & d_{12} & \cdots & d_{1N} \\ d_{21} & 0 & \cdots & d_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \cdots & 0 \end{bmatrix}$$

Each model layer has its own RDM, describing the representational geometry of the sounds at that particular layer.

Step 4: Construct the STG RDM

The same procedure is applied to the STG brain-response data. Each sound has a neural-response vector across the measured STG voxels. Pairwise Euclidean distances between these response vectors are used to construct the brain RDM:

$$\text{STG responses} \longrightarrow \mathbf{D}^{\text{STG}}$$

This RDM describes the representational geometry of the same sounds in STG.

Step 5: Compare model and brain representations

Finally, the upper-triangular entries of the model RDM and STG RDM are extracted and correlated:

$$r_l = \text{corr} \left(\text{vec}_{\text{upper}}(\mathbf{D}^{(l)}), \text{vec}_{\text{upper}}(\mathbf{D}^{\text{STG}}) \right)$$

A higher correlation indicates that the relationships between sounds at that model layer are more similar to the relationships between sounds represented in STG.

2.4 Statistical Analysis

Due to random weight initialization and stochastic training, we evaluated performance across multiple independently initialized instances of each model architecture. To make it similar to trained instances ($n = 5, 4, 5$ for Waveform, Uninspired, and Inspired models respectively), we utilized 5 random initializations for all untrained models.

For every layer of a given architecture, we computed Representational Similarity Analysis (RSA) correlation scores using Pearson or Correlation technique individually for each replicate unit. Using mean averaging, aggregated run-level scores per each layer were computed. To represent variability, 95% Confidence Intervals (CIs) were calculated across units as $\pm 1.96 \cdot \text{Standard Error of the Mean}$. To ensure unbiased testing across different architectures with different layers, we perform statistical comparisons only on the selected "peak" layers, represented by the highest mean RSA alignment with the human STG data among all layers within the trained version of that architecture. We use same estimation of peak layer for untrained models to isolate the effect of training on that specific representational space during the comparison of untrained versus trained states.

In our statistical testing we apply the Welch's two-sample t -test which is robust to unequal variances and unequal sample sizes between groups. We performed 2 types of comparisons:

1. Untrained versus Trained comparisons: We test the effect of training using ESC-50 dataset on the STG alignment. The null hypothesis assumes that there is no difference in true mean RSA scores between the trained and untrained instances at the peak layer ($\mu_{\text{trained}} = \mu_{\text{untrained}}$).
2. Architecture Comparisons: We evaluate pairwise differences across trained architectures such as Waveform, Uninspired, and Inspired models. Our null hypothesis for each separate comparison is that the true mean peak RSA scores of the two architectures are equal ($\mu_{\text{Model A}} = \mu_{\text{Model B}}$).

Traditionally to scientific analyses, we defined an alpha level of $\alpha = 0.05$. Findings are considered statistically significant if they meet $p < 0.05$ threshold.

2.5 t-SNE and UMAP Visualizations

As a complement to the quantitative RSA alignments, nonlinear dimensionality reduction was performed using t-Distributed Stochastic Neighbor Embedding (t-SNE) [3] and Uniform Manifold Approximation and Projection[4] (UMAP). Our goal is to understand how spatial topologies and semantic clusters evolve across network depths. This is an additional exploratory part to RSA rather than a replacement for formal statistical alignment.

For a comprehensive research, the peak layers of the pretrained YAMNet benchmark and a variety of layers from the custom architectures were researched. Before the dimensionality reduction, the activation tensors were flattened into one-dimensional feature vectors. This way, every plotted point in the resulting 2D space represents an activation pattern activated by a single natural sound stimulus. For visualization of all categories, we assign an unique RGB color to each category (sampled from a continuous colormap). However, with such high number of categories, some colors will be barely distinguishable, thus we also distribute 5 different geometric marker shapes (circles, squares, triangles, diamonds, and pentagons) to make the distinction between categories more clear avoiding overlapping.

For t-SNE approach we set perplexity as 30, emphasizing the preservation of local pairwise neighborhoods, while for UMAP we configure 2 hyperparameters: nearest neighbors = 15 and a minimum distance = 0.1 to balance the preservation of local clustering with global manifold

topology. For both methods actual values for hyperparameters were set heuristically, to also evaluate how good the methods will work without much configuration adjustments requiring deeper knowledge of methods. Finally, the random state is fixed at 42 in order to produce exact geometric projections across runs.

2.6 RSA Metrics Comparison

In order to check if the measured alignments are robust for to specific geometric and mathematical assumptions, we perform an evaluation of RSA metrics. We evaluate 2 stages of RSA pipeline: the distance metric used to construct the Representational Dissimilarity Matrices (RDMs), and the correlation method used to compare the model RDMs against the brain RDM. To construct pairwise distances between representations for RDMs, we apply 2 distinct metrics:

1. Euclidian distance: Measures straight-line difference between 2 vectors. This is the easiest metric which is sensitive to both the shape of activation pattern and magnitude.
2. Correlation distance: This metric normalizes the variance of each activation vector and ignores the magnitude. The later layers in networks may produce stronger reactions to sounds that often distort Euclidean geometries.

Moreover, we explore 2 different correlation coefficients which can be used to to evaluate the match between the artificial representations and the STG through RDMs:

1. Pearson correlation: It assumes a linear, proportional relationship between model feature distances and physiological brain distances.
2. Spearman Rank correlation: The method ignores actual numerical values and tests the rankings of the distances. Brain signals may scale nonlinearly compared to the mathematical distances computed in artificial neural networks.

To estimate and compare the effects of these metrics, we generate the results for all layers across all models (Waveform, Uninspired, Inspired, and YAMNet) across all four metric combinations using extracted activations. We check if the peak layer changes across geometric assumptions and finally conduct the same statistical testing using Welch’s two-sample t -test with 95% CIs as we did before. This ensures that our findings for other sections are robust and independent from artifacts of a specific distance metric.

3 Results

3.1 Effect of Training

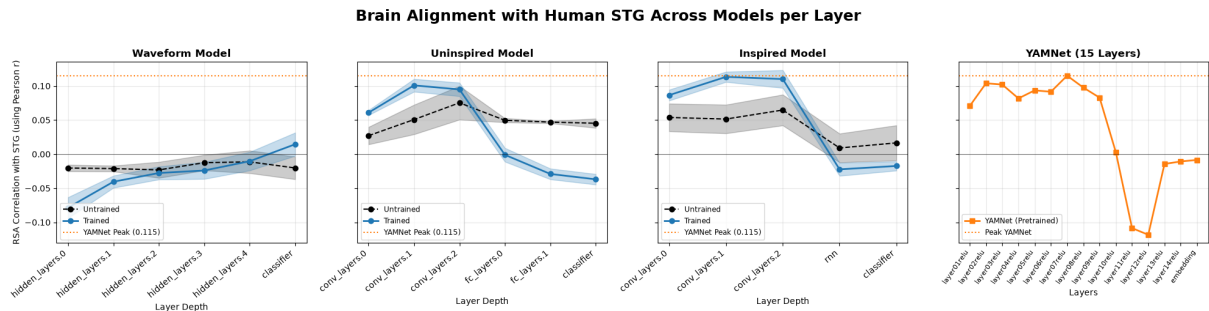


Figure 2: Layer-wise brain alignment with human STG representations for all model architectures.

Table 1: Comparison of RSA alignment between untrained and trained models at each model’s peak-performing layer.

Model	Layer	Untrained RSA (mean $\pm$ 95% CI)	Trained RSA (mean $\pm$ 95% CI)	Statistical test	<i>p</i> -value
Waveform	classifier	-0.0203 ± 0.0166	0.0145 ± 0.0170	<i>t</i> -test, <i>t</i> = 2.87	0.0208
Uninspired	conv_layers.1	0.0507 ± 0.0216	0.1009 ± 0.0093	<i>t</i> -test, <i>t</i> = 4.17	0.0073
Inspired	conv_layers.1	0.0515 ± 0.0210	0.1134 ± 0.0075	<i>t</i> -test, <i>t</i> = 5.44	0.0028

Training produced noticeable changes in all 3 architectures. For the waveform architecture, training surprisingly shifted values to more negative responses. However, the difference between trained and untrained models diminished across deeper hidden layers since the correlation metric for trained network had an upward trend while untrained one had downward trend. We can estimate that training effect was the strongest in the early feature extraction stage and weaker in later hidden layers.

The Uninspired and Inspired models had a different pattern. For both architectures training increased the measured values in all convolutional layers. In contrast, the fully connected layers and classifier shifted in the opposite direction reaching results lower than the untrained models. To quantify the effect of training, we performed *t*-tests between trained and untrained networks based on peak-performing layer determined based on trained layers. Significant differences were observed for all three models. Similar to the plot, for the Waveform models the separation between trained and untrained networks was the smallest, whereas larger effects were found for the uninspired architecture and especially for the inspired architecture. These results represent that training significantly changed the peak-layer responses of all architectures.

3.2 Effect of Architecture

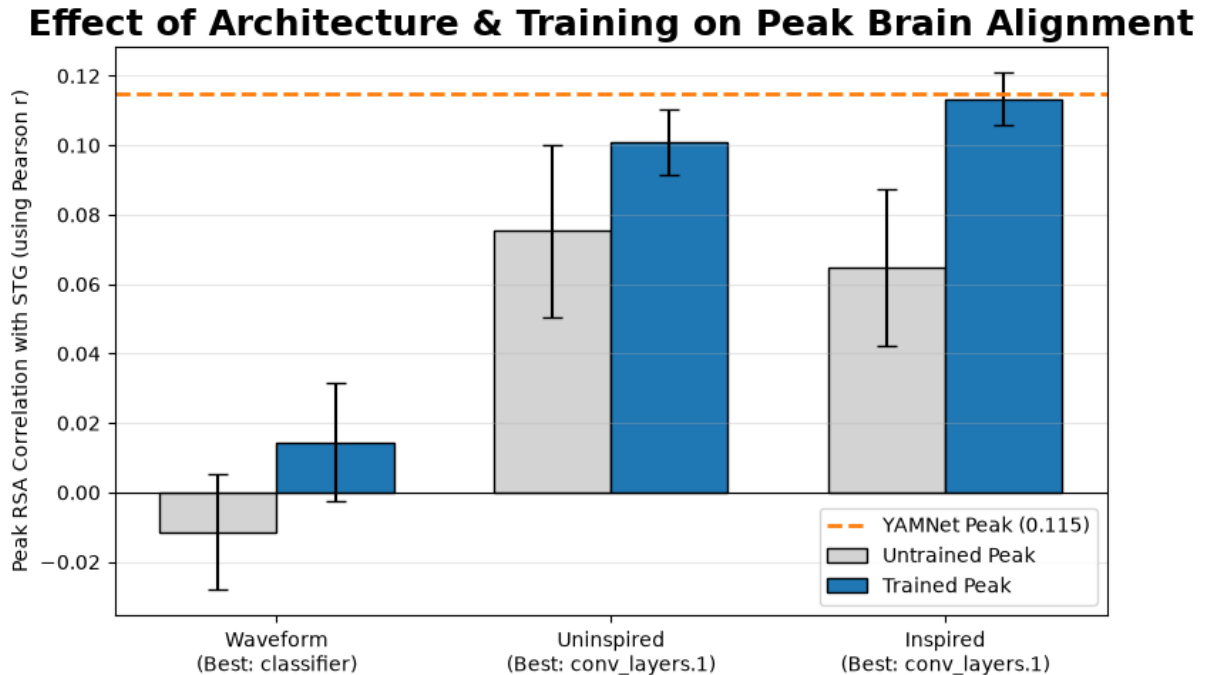


Figure 3: Effect of architecture and training on peak brain alignment.

Table 2: Peak observed brain alignment for each model family.

Model	Peak-alignment layer	Peak RSA	95% CI	Notes
WaveformModel	classifier	0.0145	± 0.0170	Weakest alignment; only positive RSA at output layer
UninspiredModel	conv_layers.1	0.1009	± 0.0093	Peak alignment in early convolutional layer
InspiredModel	conv_layers.1	0.1134	± 0.0075	Highest RSA among trained custom architectures
YAMNet	layer07relu	0.1151	N/A	Highest overall alignment; peak in intermediate layer

Table 3: Pairwise comparisons of peak brain alignment between trained model architectures.

Comparison	t-statistic	p-value
Inspired vs. Uninspired	2.04	0.0769
Inspired vs. Waveform	10.42	7.99×10^{-5}
Uninspired vs. Waveform	8.72	1.04×10^{-4}

Across architectures, peak brain alignment was mostly observed in intermediate layers rather than in final classifier layers. Even though the InspiredModel achieved a higher peak RSA (0.1134) than the UninspiredModel (0.1009), the difference appeared to be not statistically significant ($p = 0.077 > 0.05$). Both spectrogram-based architectures significantly outperformed the WaveformModel while YAMNet exhibited the highest overall alignment (RSA = 0.1151).

3.3 Effect of Model Depth

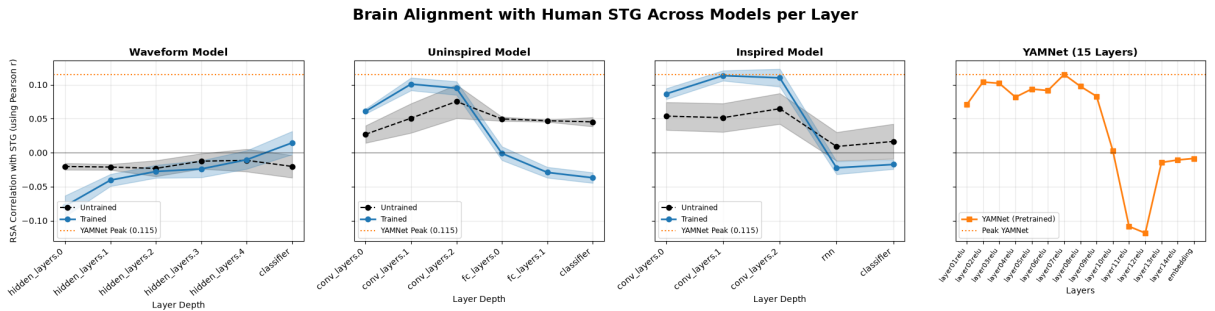


Figure 4: RSA correlation of the 4 tested models with the STG across different layers

Across the architectures we observed different trajectories, showing that the measured responses changed systematically while moving from early to later layers. For Waveform models, the correlation metric was mostly negative at the initial layers and approached 0 in deeper hidden layers. The only deviation in this pattern was the final classifier layer which exhibited a positive value.

Different from Waveform architecture but comparable to each other pattern occurred in Inspired and Uninspired models. For both models we can observe an increase across convolutional layers and then significant drop in correlation metric, becoming even negative. For both models, the highest values were observed at an intermediate depth rather than at the earliest or final processing stages.

YAMNet has more layers than other models, so we can research a more extended pattern. At first, the responses increases, reaching its maximum at layer07relu. After that, the correlation values progressively declined across deeper layers until they reached zero (0.0029), and then even becoming strongly negative (-0.1183 for layer12relu). Same drastically the metric rose towards zero and steadily converged, reaching final value of -0.0085. While following the pattern of Inspired and Uninspired modules in the beginning, at the end YAMNet demonstrated additional rise, which still however was far from the best result of the model.

3.4 t-SNE and UMAP Results

Most of the plots have almost no clear shape overall, while we can see similar colors are grouped together. In general, some of the UMAP plots look more structured than the t-SNE plots.

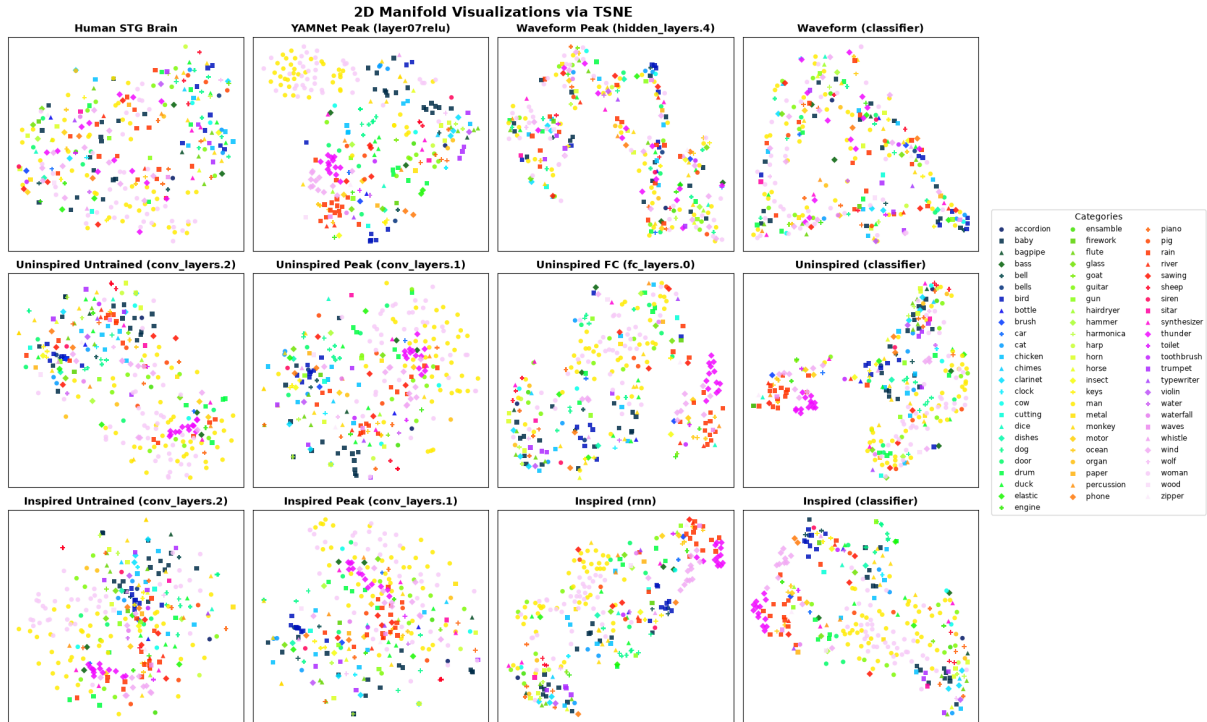


Figure 5: t-SNE visualization of auditory representations



Figure 6: UMAP visualization of auditory representations

3.5 Metrics Comparison

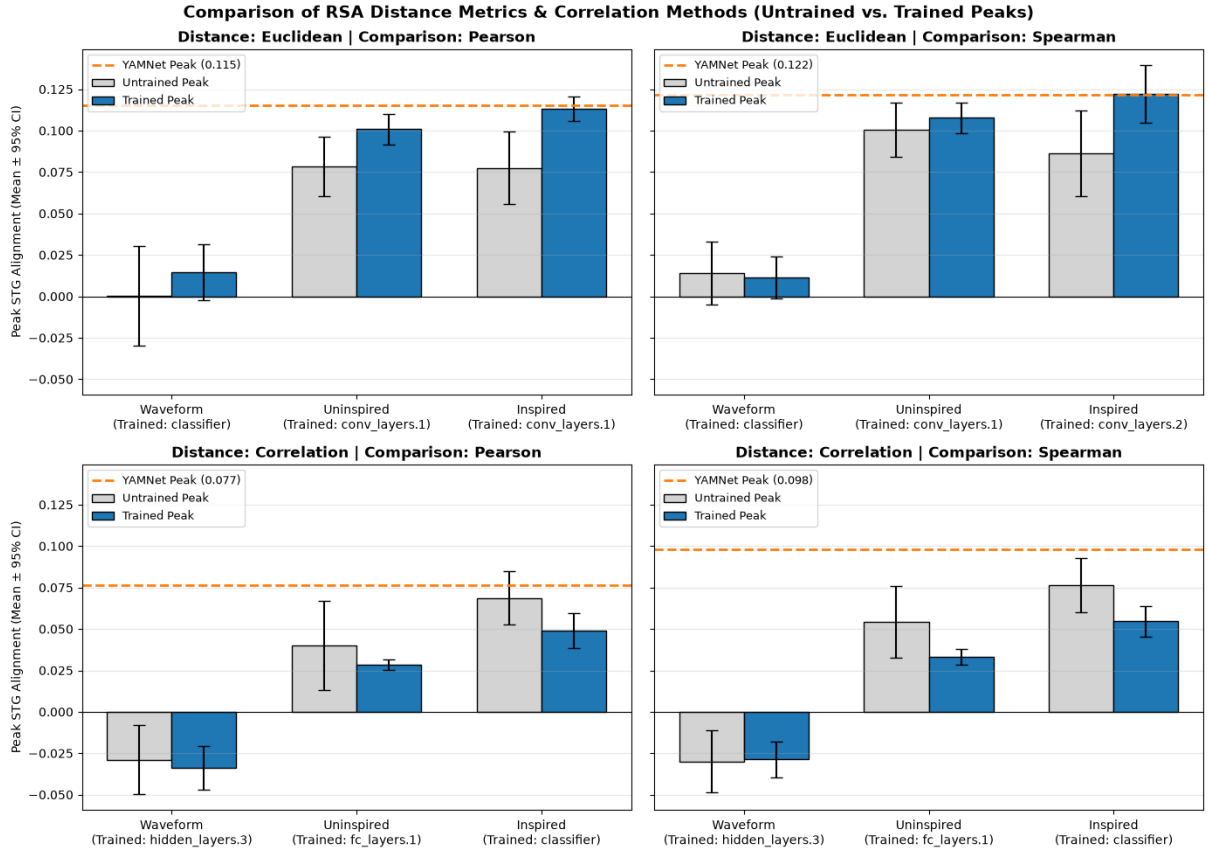


Figure 7: Comparison of RSA metrics

To assess whether the previous findings depended on the specific choice of similarity metrics, we additionally examined three alternative combinations: Euclidean distance with Spearman correlation, correlation distance with Pearson correlation, and correlation distance with Spearman correlation. The general pattern of the results remained the same: the Inspired architecture continued to outperform the Uninspired one after training, and the pretrained YAMNet model remained among the strongest-performing networks. However, Euclidian-based RDMs consistently got the scores higher than for correlation-based approach with no difference for correlation method used. It shows that the effects were influenced not by a particular correlation measure, but by the choice of distance metric. Still, overall we can say that the pattern is confirmed across all variety of approaches used, so previous results and conclusions are robust, while highlighting Euclidean distance as the most suitable metric for measuring the structure in the present dataset.

3.5.1 Correlation vs Euclidian Test

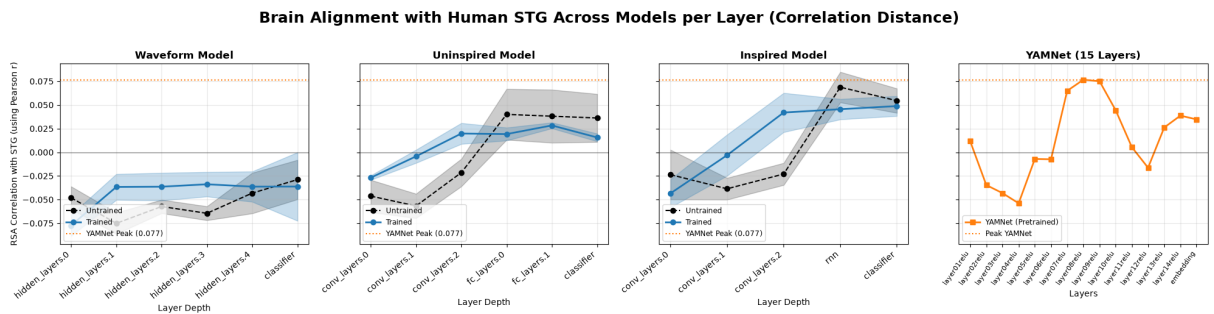


Figure 8: RSA correlation of the 4 tested models with the STG across different layers

Table 4: Comparison of RSA alignment between untrained and trained models.

Model	Layer	Untrained RSA (mean $\pm$ 95% CI)	Trained RSA (mean $\pm$ 95% CI)	Statistical test	<i>p</i> -value
Waveform	hidden_layers.3	-0.0645 ± 0.0074	-0.0338 ± 0.0130	<i>t</i> -test, $t = 4.03$	0.0061
Uninspired	fc_layers.1	0.0381 ± 0.0280	0.0283 ± 0.0031	<i>t</i> -test, $t = -0.68$	0.5343
Inspired	classifier	0.0545 ± 0.0129	0.0488 ± 0.0105	<i>t</i> -test, $t = -0.67$	0.5233

We decided to further explore the Correlation distance approach because it showed different results from Euclidian distances compariosn. For that, we conducted the same analysis as we did before. Based on new results we can observe significant decrease in peak RSA alignment for all architectures. Moreover, in a new setting only Waveform model showed a statistically significant improvement following training while still remaining negative correlation. For other architectures the strong training effects observed with Euclidean distance were not reproduced. The choice of distance measure also changed the location of the peak RSA alignment within each network as well as the trajectory across layers. This probably indicates that the 2 distance measures emphasize different properties, leading to stronger alignment with different network stages.

4 Discussion

4.1 Effect of Training

For both the Uninspired and Inspired models, training increased the mean brain alignment at all convolutional layers. At the tested peak layer (`conv_layers.1`), this improvement was

statistically significant for both Uninspired ($t = 4.17$, $p = 0.0073$) and Inspired ($t = 5.44$, $p = 0.0028$). Therefore, training clearly improved the peak brain alignment of both convolutional architectures.

In addition, the 95% confidence intervals became narrower after training at all convolutional layers of both models. For example, the CI at `conv_layers.1` decreased from ± 0.0216 to ± 0.0093 for Uninspired and from ± 0.0210 to ± 0.0075 for Inspired. This indicates that the brain-alignment scores became more consistent across trained runs and that the estimated mean alignment is more precise.

However, this improvement does not continue into the later layers. In the trained Uninspired model, alignment decreases substantially from 0.0949 at the final convolutional layer to -0.0010 at the first fully connected layer and eventually -0.0369 at the classifier. Similarly, Inspired decreases from 0.1101 at the final convolutional layer to -0.0223 at the RNN and -0.0174 at the classifier. These decreases were not directly tested statistically, so they should be interpreted as substantial observed decreases rather than statistically significant differences.

One possible explanation is task specialization. The models were trained to distinguish ESC-50 sound classes, so their later layers may become increasingly specialized for this classification objective. While the intermediate convolutional representations may preserve spectrotemporal information that is also represented in STG, the later representations may transform this information into features specifically useful for classification. This could make their representational geometry less similar to the Santoro STG responses. Overfitting could potentially contribute as well, but the current experiment does not provide enough evidence to distinguish overfitting from normal task specialization.

The Waveform architecture behaves very differently. Almost all hidden layers show weak negative correlations with STG both before and after training, and several become even more negatively aligned after training. The classifier is an exception, increasing from -0.0203 to 0.0145 . The trained-untrained difference at the classifier was statistically significant ($t = 2.87$, $p = 0.0208$), although the absolute correlation remains very small. Therefore, training does change the Waveform representation, but it does not produce the strong positive STG alignment observed in the convolutional architectures.

Before training, there is no consistent advantage of Uninspired or Inspired. Inspired has higher alignment at the first convolutional layer (0.0538 vs. 0.0271), they are almost identical at the second layer (0.0515 vs. 0.0507), while Uninspired is higher at the third layer (0.0753 vs. 0.0647). After training, however, Inspired has higher mean alignment at all three convolutional layers: 0.0865 vs. 0.0609, 0.1134 vs. 0.1009, and 0.1101 vs. 0.0949. This suggests that the architectural differences of Inspired may become more relevant after learning. However, the experiment changes several architectural components simultaneously, so this effect cannot be attributed specifically to recurrence, ReLU6, or another individual component.

4.2 Effect of Architectural Choices

As discussed above, the Waveform model shows very weak STG alignment compared with the two spectrogram-based convolutional models. At the trained peak layers, Waveform reaches only $r = 0.0145$, compared with $r = 0.1009$ for Uninspired and $r = 0.1134$ for Inspired. The differences between Inspired and Waveform ($t = 10.42$, $p < 0.001$) and between Uninspired and Waveform ($t = 8.72$, $p < 0.001$) are statistically significant.

Moving from the tested raw-waveform architecture to spectrogram-based convolutional architectures is therefore associated with substantially stronger STG alignment. This suggests that spectrotemporal convolutional representations are more suitable for reproducing the represen-

tational geometry observed in STG. However, because both the input representation and the network architecture change simultaneously, we cannot determine whether the improvement is specifically caused by the spectrogram representation, convolutional processing, or their combination.

The Inspired model reaches a higher trained peak mean alignment ($r = 0.1134$) than the Uninspired model ($r = 0.1009$). However, the difference between their peak layers is not statistically significant ($t = 2.04$, $p = 0.0769$). Therefore, although Inspired has the highest mean alignment of the three provided architectures, there is not sufficient statistical evidence to conclude that its peak alignment is better than Uninspired. In particular, we cannot claim that ReLU6 or the recurrent component causes the observed difference without testing these architectural components independently.

YAMNet, being a larger pretrained audio-event classification model, reaches the highest individual brain-alignment score, peaking at `layer07relu` with $r = 0.1151$. Interestingly, this is extremely close to the peak of the much simpler Inspired model ($r = 0.1134$), a difference of only 0.0017 in the observed correlations. This suggests that greater model scale and depth do not necessarily result in proportionally greater peak STG alignment. A comparatively simple model can therefore produce an intermediate representation with a similar observed correlation to STG. However, because only one set of YAMNet activations is available and there are no confidence intervals across multiple YAMNet runs, we cannot statistically conclude that Inspired and YAMNet perform equivalently.

YAMNet also shows a similar degradation in its later layers to the pattern observed in Inspired and Uninspired. This provides additional evidence that the representation most similar to STG does not necessarily occur at the final model output.

4.3 Alignment Across Layer Depth

For the trained convolutional models and YAMNet, there is a clear pattern in which brain alignment is strongest at intermediate rather than final layers.

For trained Uninspired, alignment initially increases from 0.0609 at `conv_layers.0` to its maximum of 0.1009 at `conv_layers.1`. It remains relatively high at `conv_layers.2` (0.0949), but then drops sharply to -0.0010 at the first fully connected layer, -0.0291 at the second fully connected layer, and -0.0369 at the classifier.

A very similar pattern occurs for trained Inspired. Alignment increases from 0.0865 at `conv_layers.0` to its maximum of 0.1134 at `conv_layers.1` and remains high at `conv_layers.2` (0.1101). It then drops to -0.0223 at the recurrent layer and -0.0174 at the classifier. Interestingly, the addition of the recurrent layer therefore does not produce greater STG alignment in this experiment, despite recurrence being an important architectural difference of the Inspired model.

YAMNet shows a similar but more gradual pattern across its deeper architecture. Alignment is positive throughout layers 1–9 and reaches its maximum at `layer07relu` ($r = 0.1151$). It then decreases from 0.0828 at layer 9 to 0.0029 at layer 10, before becoming strongly negative at layers 11 (-0.1085) and 12 (-0.1183). The final embedding also shows almost no alignment with STG ($r = -0.0085$).

Together, these results suggest that STG representational geometry is captured most strongly by intermediate network representations rather than by the final representations optimized for audio-event classification. One possible interpretation is that the earliest layers primarily represent relatively low-level acoustic properties. As information moves through the convolutional layers, these low-level features are combined into more complex spectrotemporal representations, producing a representational structure more similar to STG. In the deepest layers, the models

may increasingly transform these representations toward the specific categories required by the classification task, causing their geometry to diverge from that observed in STG.

The Waveform model does not show the same clear intermediate-layer peak and remains weakly or negatively aligned throughout most of its architecture. Therefore, the intermediate-layer pattern should not be described as universal across all models. Instead, it is a consistent pattern observed in the trained spectrogram-based convolutional models and YAMNet.

4.4 Interpreting the t-SNE and UMAP Visualizations

UMAP and t-SNE both provide differing insights. Even though both they both reduce the activation space to two dimensions, they prioritize different aspects of the relationship between stimuli samples. t-SNE preserves local neighbors which results in clearly separated islands. The distances between these islands should does not generally represent the global structure of the underlying data. UMAP also prioritizes local structures, but also preserves more of the global structure of the data when compared to t-SNE.

This can be most clearly seen in the t-SNE plot for the Uninspired classifier compared to its UMAP plot where there are no clear islands and every stimuli is in a continuous structure. Consequently, we focus primarily on the UMAP plots when analyzing the similarity between the model activations and STG activations, as we consider the more continuous structure to be more relevant as meaningful comparison.

The effect of training on model activation can be most clearly seen in the UMAP plots for the Inspired model. The plot of the trained model has a clearer structure even in a layer earlier than the untrained model. The activations in the untrained model plot are distributed more uniformly, which is most easily seen by the lack of empty space that the plot of the trained model and activations of the STG clearly posses.

The effect of the depth of a layer is well visualized in the UMAP plots. The structure of the UMAP plot of the early convolution layers in both the Uninspired and Inspired models are very noisy without a well-defined structure, resembling semi-random clouds of data points. The UMAP plots for the classifiers for these models are however have very clear structures and lose the cloud-like shape of the previous layers. This is also observable in the Waveform model, but to a way lesser extent. This also might explain why the RSA correlation between the Waveform model with the STG does not change much between layers as the UMAP plots of the Waveform model, as the underlying structure in the UMAP plots do not evolve towards the STG UMAP plot structure, but also does not grow away from the more noisy STG data.

This might also give some insight of why the RSA correlation of the models go down when analyzing later layers in our models. As the UMAP plot of the STG is more noisy than the more structured plots of the classifiers. The noisier UMAP plots of the earlier convolution layers of the Uninspired and Inspired model thus look way more similar to the STG plot than the more structured plots of the classifiers.

4.5 Differences for Various Distance Metrics and Comparison Methods

The main explanation why the results between 2 different distance methods are so different is that they emphasize different aspects of activation patterns. For example, two stimuli can be far apart according to Euclidean distance while remaining similar according to correlation distance if their activation patterns have approximately the same shape but different magnitudes. We can conclude that the original training effect is metric-dependent, but still provides informative insights. Depending on what our final goal of the analysis is, the researcher should choose which

metric suits the analysis the best.

4.6 Limitations

All models were trained on a relatively small dataset, which may have limited their ability to learn robust and generalizable representations. However, we also cannot estimate the noisiness, which is also an important factor in an alignment of data and capabilities of models. We also did not have enough time to experiment extensively or complete the bonus task, which seemed really interesting.

5 Future Work

5.1 Neuroanatomically Informed Architectures

The two neuroanatomical interventions proposed are:

First, the auditory cortex, since it is connected between and within area, signals take different time to reach specific areas. In our Inspired model, only recurrence follows the convolutional stages, also alignment dropped in the GRU. We put lateral and feedback recurrence inside the convolutional stages to simulate it like biologically, so each stage receives its neighbours' timesteps later.

Second, we apply scaling of capacity. The units in one layer match relative cell counts along the auditory hierarchy.

Our hypothesis is that these biologically-sized bottlenecks result in a closer geometrical representation to STG.

5.2 Other Interventions

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We question what objective STG solves: if late-layer decline shows specialization for ESC-50 classification, a self-supervised (or multitask) objective should keep the later ones aligned. Also, we would use sensory and neuronal noise as our biological constraints for the model.

6 Conclusion

This project examined whether the training, architecture and depth affect how similarly neural network representations align with STG responses to a series of natural sounds.

We observed that training significantly increased the alignment in all three models, however the gain in the waveform was very small.

On top of that, spectrogram-based convolutional models performed notably better than the raw-waveform counterpart, while the tallest peak of the Inspired one over un-inspired was not significant enough.

YAMNet reached a peak similar to Inspired

In trained convolutional models and YAMNet, the alignment peaked at middle layers and declined after in the classification point.

Putting all of this together, these results prove that trained networks can be used as neuroconnectionist models of the auditory cortex. However, their similarity with the real structure and encoding of the brain depends on their input, architecture and layer being examined. Thus, absolute alignments remain low (peak $r = 0.11$)

References

References

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